Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals

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Formal statements

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Abstract

For the finite half-filled sine kernel, the discrepancy between its exact modular operator and the spectral scaling function has operator norm at most 7, uniformly in every even dimension. The proof combines an elementary Jacobi spectral gap with one Riccati correction whose integral vanishes. It requires the exact concentration connection but no uniform moving-edge asymptotic expansion. For fixed positive scaled Wilson mass, the longitudinal modular trace is $O_\mu(N^{-1})$ and the full modular trace is $O_\mu(\sqrt{\log N}/N)$. Separately, averaging the second moments of finite path displacements gives the nonlinear remainder $O_\mu(\sqrt{\log N/N})$. These results do not improve the boundary entropy estimate or the full relative-entropy remainder without an additional boundary argument.

1 Finite reference and exact analytic input

Let $N \geq 2$ be even and set $h = N^{-1}$. Define

$$(B_N)_{jk} = \begin{cases} 1/2, & j = k, \\ \sin[\pi(j-k)/2]/[\pi(j-k)], & j \neq k, \end{cases} \quad (J_N)_{j-1,j} = (J_N)_{j,j-1} = \frac{j(N-j)}{N},$$

for $0 \leq j, k < N$, $1 \leq j < N$, with zero Jacobi diagonal. These matrices commute, $0 < B_N < I$, and J_N has simple spectrum. Put

$$\eta(t) = \pi t \, {}_3F_2(1/4, 1/2, 3/4; 1, 3/2; 4t^2), \quad K_N = \log(I - B_N) - \log B_N, \quad E_N = K_N + N\eta(J_N/N).$$

The geometric identities are proved in [1]. We use the following exact connection, with the differential domains and spectral selection established in [2, 3, 4]. For $|t| < 1/2$, write

$$p(x) = \operatorname{sech} x, \quad q_t = 1 - 2tp, \quad s_t = \tfrac{1}{2}\sqrt{q_t}, \quad v = p^2/4.$$

The left recessive positive solution of $[-h^2\partial_x^2 + s_t^2 + h^2 v]u_t = 0$ gives

$$w_t = hu_t'/u_t, \qquad hw_t' + w_t^2 = s_t^2 + h^2 v, \qquad w_t(-\infty) = 1/2.$$

At a common mode $J_N u = \lambda u$, $B_N u = \zeta u$, $t = \lambda/N$,

$$\log \frac{\zeta}{1-\zeta} = h^{-1} \int_{\mathbb{R}} (w_{-t} - w_t)\, dx, \qquad \eta(t) = \int_{\mathbb{R}} (s_{-t} - s_t)\, dx. \tag{1}$$

For the positive differential operator above, its Green kernel satisfies

$$G_h(t; x, x) = h^{-2} \int_x^\infty \exp\left[-\frac{2}{h}\int_x^y w_t(z)\, dz\right] dy, \quad f_N'(t) = \frac{h}{2}\int_{\mathbb{R}} p(x)[G_h(t; x, x) + G_h(-t; x, x)]dx, \tag{2}$$

where $f_N(t) = \int (w_{-t} - w_t)$. These are exact identities, not truncated asymptotic formulas. The arguments below only use interior parameters; the finite spectrum will stay a controlled distance from $|t| = 1/2$.

2 A uniform modular bound

Lemma 1 (Finite Jacobi edge gap)**.** *For every $N \geq 2$, $\|J_N\| \leq (N-1)/2$.*

Proof. Take $v_i = \sqrt{\binom{N-1}{i}} > 0$. The two terms in $(J_N v)_i / v_i$ are

$$\frac{i\sqrt{i(N-i)}}{N}, \qquad \frac{(N-1-i)\sqrt{(i+1)(N-1-i)}}{N}.$$

Each square root is at most $N/2$, so their sum is at most $(N-1)/2$. A positive-vector comparison for a nonnegative symmetric matrix bounds its spectral radius by the largest such ratio. For completeness, apply $2|x_i x_j| \leq (v_j/v_i)|x_i|^2 + (v_i/v_j)|x_j|^2$ to each bond and sum. This bounds $|\langle x, J_N x\rangle|$ by $(N-1)\|x\|^2/2$. □

Lemma 2 (One integrated Riccati correction)**.** *Let $0 \leq t < 1/2$, $a = 2t$, and $\delta = 1 - a$. If $\delta \geq h$, then the discrepancy $e = \log[(1-\zeta)/\zeta] + N\eta(t)$ at a mode obeys*

$$|e| \leq h\left[\frac{2}{3\sqrt{\delta}} + \frac{3\pi}{2\delta} + \frac{2}{3} + \frac{3\pi}{4}\right]. \tag{3}$$

The same bound applies at the negative partner with t replaced by $|t|$.

Proof. For either sign, abbreviate $q = q_{\pm t}$ and set

$$a_h = \tfrac{1}{2}\sqrt{q} - \frac{hq'}{4q}.$$

For the positive parameter, $q = 1 - ap \geq \delta$ and $q'^2 \leq 2(q-\delta)$. Since $2(z-1) \leq z^3$ for $z \geq 1$, $|q'|/q^{3/2} \leq 1/\delta$. Hence $a_h \geq s_t/2$ when $\delta \geq h$. The constant lower Riccati barrier gives $w_t \geq \sqrt{\delta}/2$, and thus $w_t + a_h \geq 3\sqrt{\delta}/4$. For the negative parameter $q = 1 + ap \geq 1$, $|q'| \leq 1/2$; $h \leq 1/2$ gives $a_h \geq s_{-t}/2$ and $w_{-t} + a_h \geq 3/4$.

Direct substitution, with $\varepsilon = w_{\pm t} - a_h$, gives

$$h\varepsilon' + (w_{\pm t} + a_h)\varepsilon = h^2\left(v + \frac{q''}{4q} - \frac{5q'^2}{16q^2}\right). \tag{4}$$

All coefficients are smooth and tend to their constant limits at both ends. The solution has $\varepsilon(-\infty) = 0$. If the damping coefficient is at least $c > 0$, the integrating factor and Tonelli's theorem give

$$\|\varepsilon\|_1 \leq \frac{h^2}{c}\left\|v + \frac{q''}{4q} - \frac{5q'^2}{16q^2}\right\|_1.$$

Here $\int v = 1/2$, $|p''| \leq p$, and integration of $(q'/q)'$ (with vanishing end values) gives $\int q'^2/q^2 = \int q''/q$. For $q = 1 - ap$ the residual norm is therefore at most $1/2 + (9/16)J_\delta$, where

$$J_\delta = \int_{\mathbb{R}} \frac{ap}{1-ap}\,dx = \frac{4a}{\sqrt{\delta(1+a)}} \arctan\sqrt{\frac{1+a}{\delta}} \leq \frac{2\pi}{\sqrt{\delta}}.$$

The equality follows by $y = \tanh(x/2)$, and is continuous at $a = 0$. For $q = 1 + ap$, the same norm is at most $1/2 + 9\pi/16$, since $\int ap/(1+ap) \leq \pi$. Consequently

$$\|\varepsilon_t\|_1 \leq \frac{4h^2}{3\sqrt{\delta}}\left(\frac{1}{2} + \frac{9\pi}{8\sqrt{\delta}}\right), \qquad \|\varepsilon_{-t}\|_1 \leq \frac{4h^2}{3}\left(\frac{1}{2} + \frac{9\pi}{16}\right).$$

Finally $\int(a_h - s_{\pm t}) = -(h/4)[\log q]_{-\infty}^{\infty} = 0$. Substitution in (1), multiplication by h^{-1}, and the triangle inequality prove (3). Odd symmetry gives the negative-mode statement. □

Theorem 3 (Uniform finite-dimensional modular control)**.** *For every even $N \geq 2$,*

$$\boxed{\|E_N\| \leq 7.}$$

Moreover every mode with $|\lambda| \leq N/4$ satisfies $|e| \leq 14/N$.

Proof. Lemma 1 gives $\delta = 1 - 2|\lambda|/N \geq h$. Equation (3) is then at most $(\sqrt{2}+1)/3 + 15\pi/8 < 7$. On $|t| \leq 1/4$ one has $\delta \geq 1/2$, and its bracket is at most $2(\sqrt{2}+1)/3 + 15\pi/4 < 14$. Commutation and the simple Jacobi spectrum identify the matrix norm with the largest absolute scalar discrepancy. □

The constant is not claimed optimal. The theorem is a bound at the finite matrix modes, not a uniform expansion for arbitrary interpolation parameters approaching the endpoint.

3 Positive weighted and Wilson consequences

For the interval compression of the occupied projection of $h_m(k) = \sin k\,\sigma_x + (1 + m - \cos k)\sigma_z$, put $N = 2L$, $m = \mu/L$, and $\mu > 0$. The interval scalarization gives $A_N - B_N = S_N + D_N^{\mathrm{tr}}$. We use the established Wilson identities [5, 6]:

$$-G_N \le S_N \le G_N, \quad G_N \ge 0, \quad \operatorname{Tr} G_N \le M := \mu/\pi,$$
$$\operatorname{Tr}(G_N \mathbf{1}_{|J_N|>x}) \le a_\mu/x + b_\mu/x^2, \quad a_\mu = \sqrt{2/15}\,\mu^2/\pi, \quad b_\mu = \mu/(3\pi),$$
$$|\langle u_j, D_N^{\mathrm{tr}} u_j\rangle| \le 2\sqrt{\langle u_j, G_N u_j\rangle \zeta_j(1-\zeta_j)}.$$

No commutation of the perturbation or majorant with J_N is assumed.

Corollary 4 (Explicit weighted bounds)**.** *For every even $N \ge 2$,*

$$\operatorname{Tr}(G_N|E_N|) \le \frac{14M + 28a_\mu}{N} + \frac{112 b_\mu}{N^2}, \qquad |\operatorname{Tr}(S_N E_N)| \le \operatorname{Tr}(G_N|E_N|). \tag{5}$$

Writing $W_N = B_N(I - B_N)$ and $H_N = \sum_{j=1}^N 1/j$,

$$V_N := \operatorname{Tr}(W_N E_N^2) \le \frac{392(H_N+1)}{\pi^2 N^2} + 49 N e^{-\pi N/6}. \tag{6}$$

Hence $|\operatorname{Tr}(D_N^{\mathrm{tr}} E_N)| \le 2\sqrt{M V_N}$ and $\operatorname{Tr}[(A_N - B_N)E_N] = O_\mu(\sqrt{\log N}/N)$.

Proof. Split the reference spectrum at $|\lambda| = N/4$ and apply Theorem 3 and the majorant tail. The order inequalities bound each longitudinal diagonal expectation, giving (5). For (6), first note $s_t^2 + h^2 v \le 9/16$ for $|t| < 1/2$, $h \le 1/2$. The upper Riccati barrier is $w_t \le 3/4$; (2) then gives $G_h(t;x,x) \ge 2/(3h)$ and $f_N'(t) \ge 2\pi/3$. Since $f_N(0) = 0$, every mode has $|K_N| \ge (2\pi/3)|J_N|$ in the joint spectral calculus. Thus $\zeta_j(1-\zeta_j) \le e^{-\pi N/6}$ outside the chosen interior. The exact projection leakage identity gives

$$\operatorname{Tr} W_N = \frac{2}{\pi^2} \sum_{d \ge 1} \frac{\min(d, N)\sin^2(\pi d/2)}{d^2} \le \frac{2}{\pi^2}(H_N + 1).$$

Use $|e| \le 14/N$ on the interior and $|e| \le 7$ on its complement. The pairing inequality and Cauchy–Schwarz prove the last assertion. □

No compact-interior WKB theorem is needed for this corollary.

4 Averaged finite paths and nonlinear convergence

Let $T_N = J_N/N$, $I_p = (p!)^2/(2p+1)!$, and $P_{p,N}(q) = \sum_{i,j}(T_N^p)_{ij} e^{iq(j-i)}$.

Lemma 5 (Averaged displacement estimate)**.** *For every $N \ge 2$, odd $p \ge 1$, and real q,*

$$|P_{p,N}(q) - N I_p (2\cos q)^p| \le (4p^2 + 2) 2^{-p} |\cos q|. \tag{7}$$

There is no restriction $p < N$.

Proof. Set $\beta_+(x) = \max\{x(1-x), 0\}$. A sign sequence $\epsilon_1, \dots, \epsilon_p \in \{-1, 1\}$ has displacement $d = \sum_j \epsilon_j$ and bond offsets $b_\ell = \sum_{j<\ell} \epsilon_j + (1 + \epsilon_\ell)/2$. Its weight from start i is $w_i = \prod_{\ell=1}^p \beta_+((i + b_\ell)/N)$. Any path leaving the interval crosses a zero bond, so unrestricted path sums equal the finite matrix power, also for $p \ge N$. Since β_+ is 1-Lipschitz and at most $1/4$,

$$|w_i - \beta_+(i/N)^p| \le \frac{4^{1-p}}{N} \sum_{\ell=1}^p |b_\ell|.$$

Let c_d be the Fourier coefficients of the difference from $[\sum_i \beta_+(i/N)^p](2\cos q)^p$. They are real, symmetric in d, and supported on odd d. With uniform independent signs, $\mathbb{E} d^2 = p$ and $\mathbb{E} b_\ell^2 = \ell - 1/2$. Summing the weight error over starts and paths therefore gives

$$\sum_d |d|\,|c_d| \le 4^{1-p} 2^p \sum_{\ell=1}^p \sqrt{p(\ell - 1/2)} \le 4p^2 2^{-p}.$$

Pair d with $-d$ and use $|\cos(dq)| \le |d||\cos q|$ for odd d. Finally the total variation of β_+^p is $2\cdot 4^{-p}$; its Riemann-sum error, multiplying the entire $(2\cos q)^p$, is at most $2^{1-p}|\cos q|$. Adding proves (7). □

Put $F(t) = \eta(t) - \pi t = \sum_{n\ge 1} a_n t^{2n+1}$, where

$$a_n = \frac{\pi\binom{2n}{n}\binom{4n}{2n}}{(2n+1)16^n}, \qquad a_n 2^{-(2n+1)} \le \frac{1}{2\sqrt{2}n(2n+1)}.$$

For the Wilson model, let $\mathcal{Q}_N = -N\,\mathrm{Tr}[(S_N + D_N^{\mathrm{tr}})F(T_N)]$ and $c_\eta = 11/18 - \pi/12 - (\log 2)/2$. The exact longitudinal Toeplitz symbol and angular average are

$$s_r(q) = -\operatorname{sgn}(\cos q)g_r(q), \quad g_r(q) = \frac{1}{2}\left(1 - \frac{|\cos q|}{\sqrt{\cos^2 q + r^2\sin^2 q}}\right), \quad r = \frac{\mu}{N+\mu}, \quad \langle u\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} u(q)\,dq.$$

Set $J_1(r) = \langle g_r|\cos q|\rangle$. For $0 < r \le 1/2$,

$$J_1(r) \le \frac{r^2}{\pi}\left(1 + \frac{1}{2}\log\frac{2}{r}\right).$$

The Toeplitz trace identity and (7) bound the degree-p trace error by $N(4p^2+2)2^{-p}J_1(r)$. For $p = 2n+1$,

$$a_n(4p^2+2)2^{-p} \le \frac{4p + 2/p}{2\sqrt{2}n} \le \frac{19}{3\sqrt{2}}.$$

Thus truncating through $n = K$ costs at most $19NJ_1(r)K/(3\sqrt{2})$.

Theorem 6 (Nonlinear remainder)**.** *For fixed $\mu > 0$ and even $N \to \infty$,*

$$\boxed{\mathcal{Q}_N = \mu^2 c_\eta + O_\mu(\sqrt{\log N/N}).}$$

Proof. We state the remaining established bounds explicitly to separate them from the new path estimate. Let $C_\eta = 8F(1/2)$, $M_{2,N} = \mathrm{Tr}(J_N^2 W_N) \le 32(H_N+1)/\pi^2$, and

$$J_3(r) = \left\langle |\cos q|^3\left(\frac{r^2\sin^2 q}{4\cos^2 q} - g_r(q)\right)\right\rangle \le \frac{r^4}{\pi}\left(\frac{1}{2} + \frac{3}{8}\log\frac{2}{r}\right),$$

with continuous interpretation at $\cos q = 0$. The scalar expansion, mass normalization, limiting-series tail, positive-majorant tail and transverse fourth moment from [7, 8, 9], combined with the path sum above, give for even $N \ge \max(2,\mu)$ and every integer $K \ge 1$:

$$\begin{aligned}|\mathcal{Q}_N - \mu^2 c_\eta| \le{}& \frac{19}{3\sqrt{2}}NJ_1(r)K + \frac{2C_\eta}{35}N^2J_3(r) + \frac{2c_\eta\mu^3}{N} + \frac{\mu^2}{384K^3}\\ &+ \frac{\sqrt{2}}{K}\left(\frac{3a_\mu}{8} + \frac{3b_\mu}{2N}\right) + 2C_\eta\sqrt{\left(\frac{a_\mu}{6N} + \frac{b_\mu}{2N^2}\right)M_{2,N}}.\end{aligned} \tag{8}$$

These are the six terms of the quantitative nonlinear proof in [7], with only its path estimate replaced by Lemma 5; each other bound is uniform in N, K. Take $K = \lfloor\sqrt{N/\log N}\rfloor$. The first and fifth terms are $O_\mu(\sqrt{\log N/N})$, the transverse term has the same order, and the other terms are smaller. This proves the result. □

5 Scope and verification

The uniform norm and weighted modular estimates use the exact connection, finite Jacobi comparison, and the stated Wilson majorant and pairing. The nonlinear refinement uses finite path second moments; no cancellation between signed path weights is presumed. Neither rate is claimed optimal. The boundary entropy contribution remains separate, so these results alone do not sharpen the full $O_\mu(1/\log L)$ relative-entropy remainder. The fixed-mass results have a conditional bounded-mass additive extension; see Calibration, identified sets and uniform limits, Proposition 7. Relative vanishing-mass and general-filling limits remain separate.

Exact rational path computations and symbolic substitution check the finite formulas. Multiprecision finite-matrix and scalar-integral checks are numerical diagnostics, not premises of these proofs or interval certificates. No claim of priority or external peer review is made.

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